Loop Unroll and Schedule Example 3

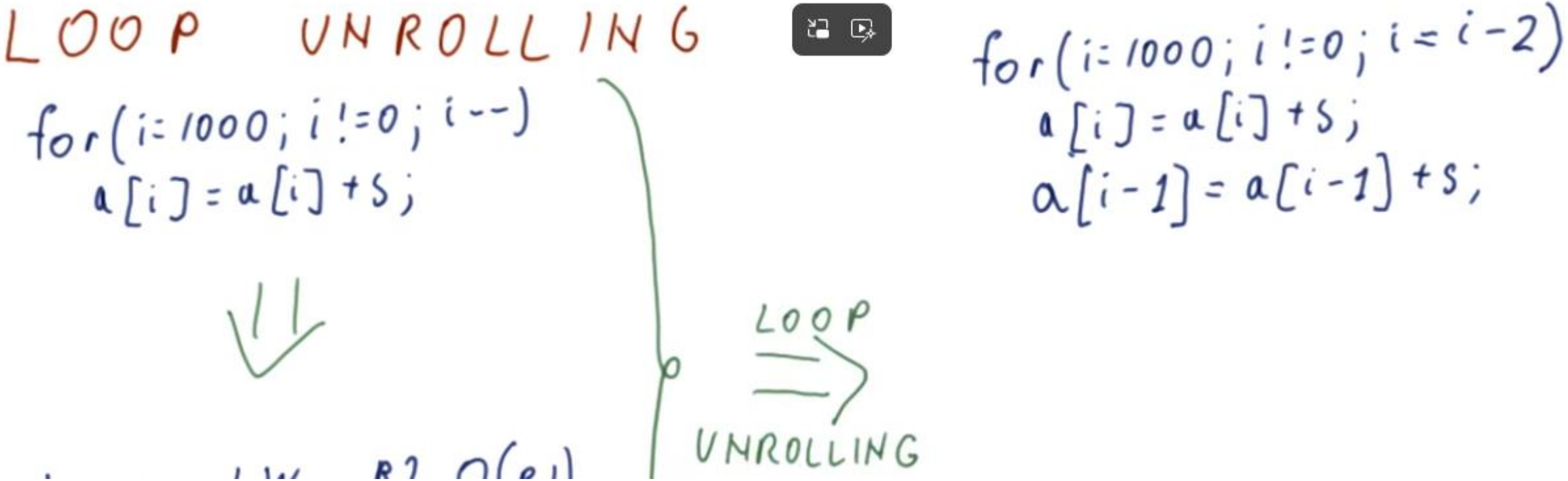
https://www.youtube.com/watch?v=EzDGaxZFU8M

LOOP UNROLLING

```
for(i=1000; i!=0; i--)
    a[i] = a[i] + s;
```



```
Loop: LW   R2, 0(R1)
      ADD  R2, R2, R3
      SW   R2, 0(R1)
      ADDI R1, R1, -4
      BNE  R1, R5, Loop
```



Loop Unrolling - Georgia Tech - HPCA: Part 3

LOOP UNROLLING

```
for(i=1000; i!=0; i--)
    a[i] = a[i] + s;
```



```
Loop: LW   R2, 0(R1)
      ADD  R2, R2, R3
      SW   R2, 0(R1)
      ADDI R1, R1, -4
      BNE  R1, R5, Loop
```

LOOP UNROLLING

Unrolled ONCE

```
for(i=1000; i!=0; i=i-2)
    a[i] = a[i] + s;
    a[i-1] = a[i-1] + s;
```



```
Loop: LW   R2, 0(R1)
      ADD  R2, R2, R3
      SW   R2, 0(R1)
      LW   R2, -4(R1)
      ADD  R2, R2, R3
      SW   R2, -4(R1)
      ADDI R1, R1, -8
      BNE  R1, R5, Loop
```

LOOP UNROLLING BENEFITS: # Insts ↓

```
Loop: LW   R2,0(R1)
      ADD  R2,R2,R3
      Sw   R2,0(R1)
      ADDI R1,R1,-4
      BNE  R1,R5,Loop
```

#Insts = 5 x 1000 = 5000

LOOP UNROLLING ⇒

```
Loop: LW   R2,0(R1)
      ADD  R2,R2,R3
      Sw   R2,0(R1)
      LW   R2,-4(R1)
      ADD  R2,R2,R3
      Sw   R2,-4(R1)
      ADDI R1,R1,-8
      BNE  R1,R5,Loop
```

#Insts = 8 x 500 = 4000

ExeTime = #Insts x CPI x CycleTime

INSTRUCTION SCHEDULING

```
LOOP:
    LW    R2, 0(R1)
    ADD   R2, R2, R0
    SW    R2, 0(R1)
    ADDI  R1, R1, 4
    BNE   R1, R3, LOOP
```

LW = 2 cycles
ADD = 3 cycles

```
LW    R2, 0(R1)
STALL
ADD   R2, R2, R0
STALL
STALL
SW    R2, 0(R1)
ADDI  R1, R1, 4
STALL
STALL
BNE
```

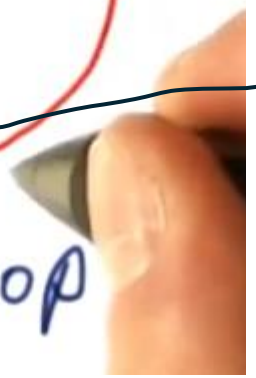


10 Cycles

INSTRUCTION SCHEDULING

```
LOOP:
    LW    R2, 0(R1)

    ADD   R2, R2, R0
    SW    R2, 0(R1)
    ADDI  R1, R1, 4
    BNE   R1, R3, LOOP
```



INSTRUCTION SCHEDULING

```
LOOP:
    LW    R2, 0(R1)
    ADDI  R1, R1, 4
    ADD   R2, R2, R0
    SW    R2, -4(R1)
    BNE   R1, R3, LOOP
```



INSTRUCTION SCHEDULING

```
LOOP:
    LW    R2, 0(R1)
    ADDI  R1,R1,4
    ADD   R2, R2, R0
    SW    R2,-4(R1)
    BNE   R1, R3, LOOP
```

→

```
LW    R2, 0(R1)
ADDI  R1, R1, 4
ADD   R2, R2, R0
STALL
STALL
SW    R2, -4(R1)
```

→

→

→

```
BNE   R1, R3, LOOP
```

LOOP UNROLLING QUIZ

```
Loop:
  LW   R1, 0(R2)
  ADD  R3, R3, R1
  ADDI R2, R2, 4
  BNE  R2, R4, Loop
```



LOOP UNROLLING QUIZ

```
Loop:
  LW   R1, 0(R2)
  ADD  R3, R3, R1
  ADDI R2, R2, 4
  BNE  R2, R4, Loop
```

- IN-ORDER, 2 INST/CYCLE
- LW: 3 CYCLES
- ADD/ADDI: 2 CYCLES

AFTER SCHEDULING

_____ CYCLES/1000 ITERATION

AFTER UNROLLING ONCE AND SCHEDULING

_____ CYCLES/1000 ITERATION